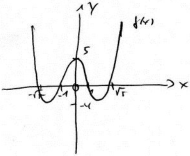
1\. a) $f(x) = x^4 - 6x^2 + 5$ Achsensymmetrie, nur gerade Exp.   Pass F13



$f'(x) = 4x^3 - 12x$

$f''(x) = 12x^2 - 12$

NST: $(x^2-5)(x^2-1) = 0$

$x_{1,2} = \pm\sqrt{5}$

$x_{3,4} = \pm 1$

Extr.: $4x(x^2-3) = 0$

$x_1 = 0$ $y_1 = 5$ $f''(0) < 0 \Rightarrow \text{Max}(0|5)$

$x_{2,3} = \pm\sqrt{3}$ $y_{2,3} = -4$ $f''(\pm\sqrt{3}) > 0 \Rightarrow \text{Min}(\pm\sqrt{3}|-4)$

WP $12(x-1)(x+1) = 0$

$x_{1,2} = \pm 1$ $y_{1,2} = 0$ einfache NST, Vorzeichenwechsel, WP$(\pm 1, 0)$

Wertebereich: $W = [-4; \infty[$ wegen Minima und Achsensymmetrie

b) $g(x) = x^2 + 5$ $f = g$

$g'(x) = 2x$ $x^2(x^2-7) = 0$

$x = 0$

$x = \pm\sqrt{7}$

$\tan\alpha_f = f'(\sqrt{7}) = 16\sqrt{7}$

$\alpha_f = 88{,}64°$

$\underline{\alpha = \alpha_f - \alpha_g = 9{,}35°}$

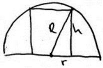
$\tan\alpha_g = g'(\sqrt{7}) = 2\sqrt{7}$

$\alpha_g = 79{,}298°$

c) $A = \int_{-\sqrt{7}}^{\sqrt{7}} (g-f)\,dx = \underbrace{2\int_0^{\sqrt{7}} (g-f)\,dx}_{\text{Symmetrie}} = 2 \cdot \int_0^{\sqrt{7}} (-x^4 + 7x^2)\,dx$

$= 2\left[-\frac{1}{5}x^5 + \frac{7}{3}x^3\right]_0^{\sqrt{7}} = \underline{\frac{196}{15}\sqrt{7}}$

2. Pass F13



$R^2 = h^2 + r^2$

$r^2 = R^2 - h^2$

$V = r^2 \pi h = (R^2 - h^2)\pi h = \pi(R^2 h - h^3)$ $\qquad D_h = [0; R]$

$V' = \pi(R^2 - 3h^2) = 0$

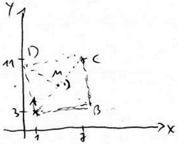
$h = (\pm)\frac{1}{3}\sqrt{3}R$

Ränder D: $V(0) = 0$

$V(R) = 0$

$V(h) = \frac{2}{9}\pi\sqrt{3}\, r^3$ $\qquad$ 58% Halbkugelvolumen

3.



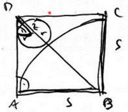
$$\vec{r}_M = \frac{\binom{1}{3}+\binom{7}{11}}{2} = \binom{4}{7} \qquad M(4|7)$$

$$\vec{AC} = \binom{6}{8} \qquad \vec{AM} = \frac{1}{2}\vec{AC} = \binom{3}{4} \xrightarrow{\text{D}} \vec{MB} = \binom{4}{-3}$$

$$\vec{MD} = \binom{-4}{3}$$

$$\vec{r}_B = \vec{r}_M + \vec{MB} = \binom{4}{7} + \binom{4}{-3} = \binom{8}{4} \qquad \underline{B(8|4)}$$

$$\vec{r}_D = \vec{r}_M + \vec{MD} = \binom{4}{7} + \binom{-4}{3} = \binom{0}{10} \qquad \underline{D(0|10)}$$



$$\frac{DZ}{r} = \frac{DB}{s} \qquad DZ = s\cdot\sqrt{2} - s - r$$

$$DB = s\sqrt{2}$$

$$\frac{s\sqrt{2}-s-r}{r} = \frac{s\sqrt{2}}{s}$$

$$s(\sqrt{2}-1) = r(1+\sqrt{2})$$

$$r = \frac{\sqrt{2}-1}{\sqrt{2}+1}s$$

$$\underline{r = (3-2\sqrt{2})s}$$

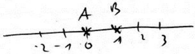
$$\vec{DZ} = \quad r \qquad \vec{BZ} = \frac{s+r}{s\sqrt{2}}\cdot\vec{BD} = \frac{(4-2\sqrt{2})s}{s\sqrt{2}}\cdot\binom{-8}{6} = (2\sqrt{2}-2)\binom{-8}{6}$$

$$\vec{r}_Z = \vec{r}_B + \vec{BZ} = \binom{8}{4} + (2\sqrt{2}-2)\binom{-8}{6} = 4\binom{6-4\sqrt{2}}{-2+3\sqrt{2}} \approx \binom{5{,}5}{3{,}6}$$

$$\underline{Z(24-16\sqrt{2}\,|\,-8+12\sqrt{2})}$$

Pass F13

4.



A, B pro Sekunde ± 1, $p = 50\%$

a) P(A nach einer Sekunde rechts von B) : A nach rechts, B nach links

$= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 25\%$

b) P(" zwei " ") :

je $(\frac{1}{2})^2 \cdot (\frac{1}{2})^2 = \frac{1}{16}$

$A \to 2$: 1
$B \to 1, 0, -1$: 3 } 3
2 – 1

$A \to 1$: –

$A \to 0$: 2
$B \to -1$, 1 } 2

$A \to -1, -2$ ↯

5

$= 5 \cdot \frac{1}{16} = \frac{5}{16} = 31{,}3\%$

c) P(A, B nach 10 Sek. am gleichen Platz)

d.h. jeder 5 mal nach rechts, 5 mal nach links $\binom{10}{5} = 252$

$= 252 \cdot (\frac{1}{2})^{10} \cdot 252 \cdot (\frac{1}{2})^{10} = 6{,}1\%$

5. $f(x) = 2e^{-x} - 1$

Pass F13

a) $f(0) = \underline{1 = y}$

$2e^{-x} - 1 = 0$

$e^{-x} = \frac{1}{2}$

$\underline{x} = -\ln\frac{1}{2} = \underline{\ln 2}$

$f'(x) = -2e^{-x}$

$f'(\ln 2) = -2e^{-\ln 2} = -1 = \tan\alpha \Rightarrow \underline{\alpha = -45° = 135°}$

$$A = \int_0^{\ln 2} f(x)\,dx = \left[-2e^{-x} - x\right]_0^{\ln 2} = -2e^{-\ln 2} - \ln 2 + 2$$

$\underline{A = 1 - \ln 2}$

b) $g(x) = ae^{-x} - 1$ $\qquad e^{-x} = \frac{1}{a}$

$g'(x) = -ae^{-x}$ $\qquad x = \ln a$

$g'(\ln a) = -ae^{-\ln a} = \frac{-a}{e^{\ln a}} = \frac{-a}{a} = -1 = \tan\alpha$

$\underline{\alpha = -45° = 135°}$

g^{-1}: $x = a \cdot e^{-y} - 1$

$e^{-y} = \frac{1+x}{a}$

g^{-1}: $y = -\ln\frac{1+x}{a} = \ln\left(\frac{a}{1+x}\right)$

$g^{-1}(0) = 2 = \ln\left(\frac{a}{1+0}\right) = \ln a$

$\underline{e^2 = a}$

oder: Umkehrfunktion durch $(0|2)$ bedeutet $(2|0)$ auf Funktion:

$g(2) = 0 = ae^{-2} - 1$

$\underline{e^2 = a}$